Damien Robert

Deep Learning for Remote Sensing & Environment

damien.robert@uzh.ch ▪ drprojects.github.io ▪ drprojects ▪ 0000-0003-0983-7053 ▪ Scholar ▪ damienrobert91

I am a postdoctoral researcher in the EcoVision lab at University of Zurich, I collaborate with Jan Dirk Wegner to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on *"Efficient Learning on Large-Scale 3D Point Clouds"* at IGN and ENGIE, under the supervision of Loic Landrieu and Bruno Vallet.

POSITIONS

2024–now **Postdoctoral Researcher**
EcoVision lab, University of Zurich · Zurich, Switzerland
Deep learning for remote sensing and environment
PI: Jan D. Wegner

2020–2024 **PhD Student**
ENGIE Lab CRIGEN — LASTIG, IGN/ENSG · Paris, France
Efficient learning on large-scale 3D point clouds
Advisors: Loic Landrieu, Bruno Vallet

2017–2020 **R&D Engineer**
SIRADEL, ENGIE · Rennes, France
Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data

2017 **Co-Founder**
Inspirama
Website gathering book recommendations from inspiring people

2015 **R&D Intern**
Dassault Systèmes
Dimensionality reduction and dynamic system modeling

EDUCATION

2011–2015 **MSc, Engineering**
École Centrale de Lyon · Lyon, France
Mathematics, Computer Science, Mechanics, Signal Processing, Automation

2022 **International Computer Vision Summer School**
ICVSS · Sicily, Italy
Computer vision courses by world-renowned experts in academia and industry

2017 **CNRS AI Fall School**
CNRS · Lyon, France
Multi-disciplinary course for AI students and researchers

AWARDS

2026 **Outstanding Reviewer** · CVPR

2025 **PhD Award — accessit** · AFRIF
French Association for Pattern Recognition and Interpretation

2024 **Outstanding Reviewer** · ECCV

2022 **Best Paper Finalist** · CVPR
Top 0.4% of submissions

PUBLICATIONS

2022–2026

2026

Bias Aware Benchmark for Species Distribution Modeling
E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, **D. Robert**, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia
World Biodiversity Forum

Opportunities and challenges for near-term ecological forecasting with biodiversity and EO data
P. Singh, B. Bektas, S. Evers, E. Fenollosa, G. Koren, S. K. M. Parvathi, S. Pang, M. Paniw, **D. Robert**, G. Sialelli, J. Slingsby, R. Thornley, E. L. Underwood, J. D. Wegner, W. Wu
World Biodiversity Forum

LitePT: Lighter Yet Stronger Point Transformer
Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler
CVPR · Paper · Project · Code — Highlight

EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation
L. Geist, L. Landrieu, **D. Robert**
ICRA · Paper · Project · Code

FORMSpoT: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series
M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciais
Remote Sensing of Environment · Paper

2025

MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation
K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner
PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science · Paper · Code

SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology
E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann
CVPR workshop EarthVision · Paper · Project · Code

GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance
K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner
ISPRS Geospatial Week · Paper · Project · Code — Oral

Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks
J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner
ICLR workshop Tackling Climate Change with Machine Learning · Paper · Project · Code

Lossy Neural Compression for Geospatial Analytics: A Review
C. Gomes, I. Wittmann, **D. Robert**, J. Jakubik, T. Reichelt, M. Martone, S. Maurogiovanni, R. Vinge, J. Hurst, E. Scheurer, R. Sedona, T. Brunschwiler, S. Kesselheim, M. Batic, P. Stier, J. D. Wegner, G. Cavallaro, E. Pebesma, M. Marszalek, M. A. B. Plomer, K. Adriko, P. Fraccaro, R. Kienzler, R. Briq, S. Benassou, M. Lazzarini, C. M. Albrecht
IEEE Geoscience and Remote Sensing Magazine · Paper

2024

Efficient Learning on Large-Scale 3D Point Clouds
D. Robert
PhD Thesis, Université Gustave Eiffel
Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet ·
Paper · Code — AFRIF award

Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering
D. Robert, H. Raguet, L. Landrieu
3DV · top 5.3% of submissions · Paper · Project · Code — Oral

2023

Efficient 3D Semantic Segmentation with Superpoint Transformer
D. Robert, H. Raguet, L. Landrieu
ICCV · top 26.8% of submissions · Paper · Project · Code

2022

Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation
D. Robert, B. Vallet, L. Landrieu
CVPR · top 0.4% of submissions · Paper · Project · Code — Oral · Best paper finalist

UNDER REVIEW

2026 **Toward a Foundation Model for Forest Point Clouds**
Y. Yue, S. Puliti, **D. Robert**, A. Topaloğlu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler

2026 **From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps**
G. Sialelli, R. Young, Y. Jiang, C. Aybar, L. Scheibenreif, **D. Robert**, C. Mosig, A. J. Stewart, J. D. Wegner, A. Pirinen, O. Mogren, K. Schindler

COLLABORATION & SUPERVISION

PhD Students: **Emilia Arens** UZH with Jan D. Wegner · **Johannes Dollinger** UZH with Jan D. Wegner · **Kaan Karaman** UZH with Jan D. Wegner · **Louis Geist** ENPC with Loic Landrieu · **Yuanwen Yue** ETH / Oxford with Konrad Schindler and Christian Rupprecht

Master Students: **Valerio Schelbert** ETH · **Jackson Sunny** École Polytechnique with SAMP

Collaborators: **Konrad Schindler** ETH · **Christian Rupprecht** Oxford · **Jan D. Wegner** UZH · **Loic Landrieu** ENPC · **Vivien Sainte Fare Garnot** UZH · **Lukas Drees** UZH · **Martin Schwartz** LSCE · **Elena Plekhanova** WSL

Advising: **UpCircle AI** Switzerland · **École des Sciences Criminelles de Lausanne** Switzerland · **SAMP AI** France · **GEODIGITAL** US

REVIEWING & CHAIRING

2026	ECCV area_chair · CVPR reviewer
2025	CVPR reviewer · CVPR workshop EarthVision reviewer
2024	ECCV reviewer · ISPRS Journal of Photogrammetry and Remote Sensing reviewer · ICLR workshop ML4RS reviewer · CVPR workshop EarthVision reviewer
2023	CVPR reviewer · CVPR workshop EarthVision reviewer
2022	ISPRS Journal of Photogrammetry and Remote Sensing reviewer · CVPR workshop EarthVision reviewer

TEACHING

2026	**Transformer** · UZH · Lecture, Master 2	
2025	**AI4Good group projects** · UZH · Labs, Master 2	14 h
2025	**Transformer** · UZH · Lecture, Master 2	4 h
2024	**Transformer, NeRFs, and Diffusion** · UZH · Lecture and labs, Master 2	5.5 h
2023	**Deep Learning for Remote Sensing** · ENSG-IGN · Lecture, Master 2	13 h
2022	**3D Deep Learning for Remote Sensing** · ISPRS'22 · Tutorial, Researchers	1 day
2022	**3D Deep Learning, Torch-Points3D & DeepViewAgg** · ENGIE CRIGEN lab · Tutorial, Researchers	1 day
2022	**Deep Learning for Remote Sensing** · ENSG-IGN · Lecture, Master 2	9 h
2020	**Deep Learning for Computer Vision** · Ecole Polytechnique · Lecture, Master 1	12 h

OPEN-SOURCE SOFTWARE

LitePT · prs-eth/LitePT	405	44
torch-graph-components · drprojects/torch-graph-components	15	0
Superpoint Transformer · drprojects/superpoint_transformer	1054	136
NoRA · drprojects/nora	53	3
Point Geometric Features · drprojects/point_geometric_features	74	8
DeepViewAgg · drprojects/DeepViewAgg	237	24

SELECTED TALKS

2026 **ESA-NASA workshop on AI Foundation Model for EO** · tutorial · Building Earth Observation Embedding Workflows with TerraTorch · Virtual

ESA Φ-lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy

2025 **MPI of Animal Behavior** · Research at EcoVision lab · Zurich, Switzerland

Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual

NatureFinance · Research at EcoVision lab · Virtual

NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland

CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual

2024 **3D Data Academy** · interview · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube

Google DeepMind · Research at EcoVision lab · Zurich, Switzerland

3DV'24 · oral · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland

2023 **Ecole des Ponts, IMAGINE lab** · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France

National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France

ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland

ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland

Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France

2022 **CV News** · interview · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US

CVPR'22 · oral · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US

45 talks given since 2021 — full list at drprojects.github.io/talks.

SKILLS & LANGUAGES

Research

3D & geometry Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning

Efficient & scalable learning Efficient deep learning · Scalable model design · EO data representation and compression

Learning paradigms Multimodal learning · Self-supervised learning · Representation learning

Environment & Earth observation Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology

Tools

Python	PyTorch	PyTorch Lightning	PyTorch Geometric
C++	Hydra	Weights & Biases	Git
Linux	SLURM	scikit-learn	Plotly
Blender	LaTeX		

Languages

French	Native	English	Fluent
Spanish	Intermediate	German	Beginner

Abroad: Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015

Interests: Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games